Technology Stack

Date	26 June 2026
Team ID	N/A
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology chosen	Justification / purpose
1	**Frontend / Client-Side** (e.g., React Angular, HTML / CSS)	Streamlit, HTML/CSS	Used to build a simple web interface for audio upload, waveform display, result viewing, and PDF download.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python	Handles audio processing, speech-to-text, semantic similarity, scoring logic, and report generation.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	Temporary File Storage	Uploaded audio files and generated reports are stored temporarily during processing and deleted after use.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Localhost / Streamlit Cloud	The app can run locally during development and can be deployed on Streamlit Cloud for public access.

5	**Version Control & CI/CD**	Git, GitHub	Used to store project code, track changes, and submit the project repository for mentor review.
	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	OpenAI Whisper, Sentence-BERT, Librosa, ReportLab, Matplotlib	Whisper converts speech to text, Sentence-BERT checks concept similarity, Librosa extracts audio features, Matplotlib creates waveforms, and ReportLab generates PDF reports.